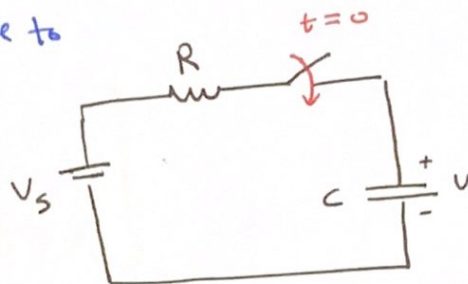


7.5

Step Response of an RC Circuits :-

The step response is the response of the circuit due to a sudden application of a dc voltage or current source.

- Capacitor voltage is the circuit response to be determined.
- Assume an initial voltage V_0 stored in the capacitor.



- Apply KCL

$$C \frac{dv}{dt} + \frac{V - V_s}{R} = 0 \Rightarrow \frac{dv}{dt} + \frac{V - V_s}{RC} = 0 \Rightarrow \frac{dv}{dt} = - \frac{V - V_s}{RC}$$

$$\frac{dv}{V - V_s} = - \frac{dt}{RC}$$

[integrate both side limited to $\int_{V=V_0}^{V=V_t}$ and $\int_{t=0}^t$]

$$\int_{V_0}^V \frac{dv}{V - V_s} = - \frac{1}{RC} \int_0^t dt \Rightarrow \ln V - V_s \Big|_{V_0}^{V_t} = - \frac{t}{RC} \Big|_0^t$$

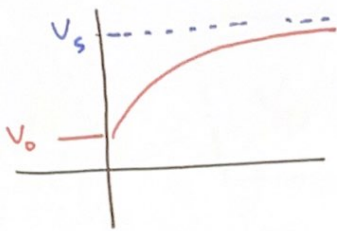
$$\ln(V_t - V_s) - \ln(V_0 - V_s) = - \frac{t}{RC} \Rightarrow \ln \frac{V_t - V_s}{V_0 - V_s} = - \frac{t}{RC}$$

$$\frac{V_t - V_s}{V_0 - V_s} = e^{-t/RC}$$

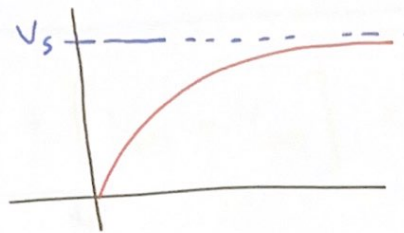
$$V_t = V_s + (V_0 - V_s) e^{-t/RC} \quad \tau = RC$$

The complete response of the RC circuit to a sudden application of a DC voltage source, assuming the capacitor is initially charged.

$$V(t) = \begin{cases} V_0 & t < 0 \\ V_s + (V_0 - V_s) e^{-t/RC} & t > 0 \end{cases}$$



initially charged capacitor

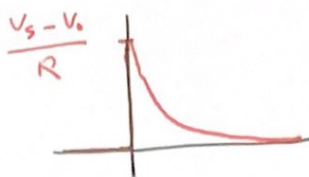


initially uncharged capacitor

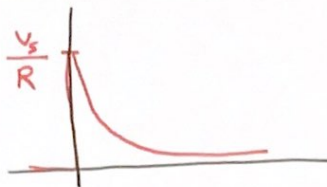
Capacitor current is obtained using $i(t) = C \frac{dv}{dt}$

$$i(t) = C \frac{d}{dt} \left[V_s + (V_0 - V_s) e^{-t/RC} \right]$$

$$i(t) = -\frac{V_0}{R} e^{-t/RC} + \frac{V_s}{R} e^{-t/RC} = \left(\frac{V_s - V_0}{R} \right) e^{-t/RC}$$



initially charged capacitor



initially uncharged capacitor

Current through capacitor gradually decline to zero as $t \rightarrow \infty$.

Complete response = transient response + steady-state response

$$V = V_t + V_{ss}$$

Where

$$V_t = (V_0 - V_s) e^{-t/\tau} \quad \leftarrow \text{as } t \rightarrow \infty \text{ } V_t \text{ die out}$$

$$V_{ss} = V_s$$

$$V(t) = V_{(\infty)} + [V_{(0)} - V_{(\infty)}] e^{-t/\tau}$$

To find the step response of an RC circuit we need:

- 1- Initial capacitor voltage V_0 .
- 2- Final capacitor voltage V_{∞} .
- 3- The time constant τ .

ex: 7.11 302

Let $V_c(0) = 60 \text{ V}$, determine V_c , V_x and i_o for $t \geq 0$

$$R_6 \parallel R_2 \Rightarrow \frac{12+6}{12+6} = 4 \Omega$$

R_4 in series with R_8

$$R_T = 4 + 8 = 12 \Omega$$

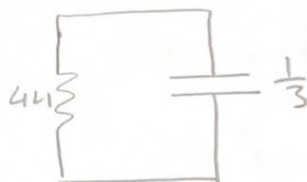
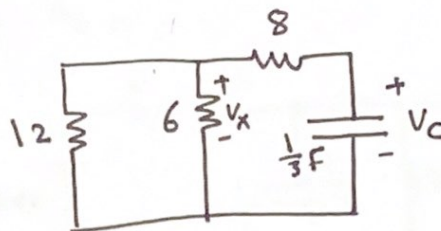
$$\tau = RC = \frac{12}{3} = 4 \text{ s}$$

$$V_t = V_0 e^{\frac{-t}{\tau}} = 60 e^{-t/4} = 60 e^{-0.25t} \text{ V}$$

$V_x = V$ on R_4

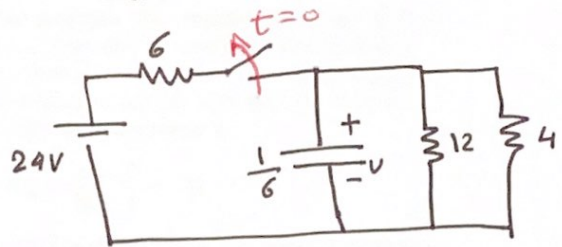
$$V_{R_4} = \frac{4}{4+8} V = \frac{1}{3} (60 e^{-t/4}) = 20 e^{-0.25t} \text{ V}$$

$$i_o = \frac{V_T}{R_T} = \frac{V}{12} = \frac{60}{12} e^{-t/4} = 5 e^{-0.25t} \text{ A (opposit)}$$



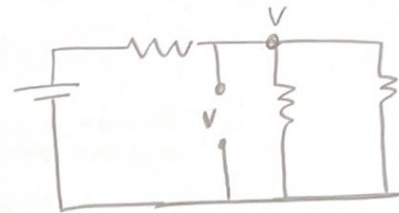
7.2 If the switch in figure below opens at $t=0$ find $V(t)$ for $t \geq 0$ and $W_C(\infty)$.

For $t < 0$ the switch is closed and at steady state the capacitor act as an O.C.



$$R_2 // R_4 = \frac{12 \times 4}{16} = 3$$

$$V = V_{R_3} = \frac{3 \times 24}{6+3} = \frac{72}{9} = 8 \text{ V}$$



Voltage across the capacitor at $t=0^-$ is same at $t=0$

$$V_C(0) = V_0 = 8 \text{ V}$$

For $t > 0$ the switch is open and the 24V is disconnected

$$\tau = R_T C = 3 \times \frac{1}{6} = \frac{1}{2} \text{ s}$$

$$V = V_0 e^{-t/\tau} = 8 e^{-2t} \text{ V}$$



or



$$W_C(\infty) = \frac{1}{2} C V_0^2 = \frac{1}{2} \times \frac{1}{6} \times 8^2 = \frac{64}{12} = 5.33 \text{ J}$$